G Explicit QE and EQ substitutions

G.1 Quotient first

On U_{reg}, $H_\perp x + J = 0$ has the unique solution (5.29). Then

$$\mathcal{L}^{\mathrm{QE}}_{\mathrm{aff,on\text{-}shell}} = \tfrac{1}{2} x_\star^{\mathsf{T}} H_\perp x_\star + J^{\mathsf{T}} x_\star. \tag{G.1}$$

Using the Schur factorization,

$$\tfrac{1}{2} x_\star^{\mathsf{T}} H_\perp x_\star + J^{\mathsf{T}} x_\star = \mathcal{L}_{\mathrm{const}} + \tfrac{1}{2} S_t t_\star^2 + J_{t,\mathrm{eff}} \cdot t_\star = \frac{3(\partial f_1)^2}{2f_1} + f_2 s^2 - (\partial f_3)\cdot s, \tag{G.2}$$

with $s = \partial f_3/(2f_2)$. The last two terms are

$$f_2\Big(\frac{\partial f_3}{2f_2}\Big)^2 - (\partial f_3)\cdot\frac{\partial f_3}{2f_2} = \frac{(\partial f_3)^2}{4f_2} - \frac{(\partial f_3)^2}{2f_2} = -\frac{(\partial f_3)^2}{4f_2}. \tag{G.3}$$

Hence

$$\mathcal{L}^{\mathrm{QE}}_{\mathrm{aff,on\text{-}shell}} = \frac{3(\partial f_1)^2}{2f_1} - \frac{(\partial f_3)^2}{4f_2}. \tag{G.4}$$

G.2 Eliminate first

Substitute (6.13) into (F.1). Write $T = s + V$, $Q = 4r - (8/3)(s+V)$, $\widetilde{Q} = r - (2/3)(s+V)$, $A = V$. Every occurrence of V in $\mathsf{Q}(T, Q, \widetilde{Q})$ and in $2T + Q - \widetilde{Q}$ cancels by Appendix C, and $T - A = s$ is independent of V. The remaining expression is a function of (r, s) only. Expanding,

$$\begin{aligned} f_1\,\mathsf{Q}\big|_{V\text{-free}} + (\partial f_1)\cdot(2s + 4r - r) + f_2 s^2 - (\partial f_3)\cdot s \\ = \frac{3(\partial f_1)^2}{2f_1} - \frac{(\partial f_3)^2}{4f_2}, \end{aligned} \tag{G.5}$$

after using $r = \partial f_1/f_1$, $s = \partial f_3/(2f_2)$, and the Palatini evaluation of Q on $(q, u) = (4r - 8s/3,\, r - 2s/3)$. Thus

$$\mathcal{L}^{\mathrm{EQ}}_{\mathrm{aff,on\text{-}shell}} = \frac{3(\partial f_1)^2}{2f_1} - \frac{(\partial f_3)^2}{4f_2}, \tag{G.6}$$

independent of V. The representatives $V = 0$, $V = \mathrm{d}\chi$, and V arbitrary therefore yield the same bulk density.

G.3 Admitted remainder

The two substitutions are performed on the bulk densities after (5.5)–(5.6). They therefore agree modulo

$$\partial_\mu\big(\sqrt{-g}\,(c_1 J_1^\mu + c_3 J_3^\mu)\big). \tag{G.7}$$

Whether this remainder is retained, canceled, or discarded is a variational-boundary convention. Off shell, $\delta J_1 = \delta J_3 = 0$ for projectively inert spectators, so the remainder does not classify proper versus improper transformations (Appendix H).

H Boundary currents and IBP identification

H.1 Currents actually used

The integrations by parts in the proof are only (5.5) and (5.6). They generate

$$J_1^\mu = f_1 W^\mu, \qquad J_3^\mu = f_3 \bar{T}^\mu, \tag{H.1}$$

with $W^\mu = -2T^\mu - Q^\mu + \widetilde{Q}^\mu$ and $\bar{T}^\mu = t^\mu = T^\mu - A^\mu$. No other surface current is invoked.